

Homework 1

Problem 1. Let $(X, \leq_1)$, $(Y, \leq_2)$ be (partially) ordered sets. We say that they are isomorphic if there exists a bijection $f : X \rightarrow Y$ such that for every $x, y \in X$, we have $x \leq_1 y$ if and only if $f(x) \leq_2 f(y)$.

1. Draw Hasse diagrams for all nonisomorphic 3-element posets.
2. Prove that any two n -element linearly ordered sets are isomorphic.
3. Prove that $(\mathbb{N}, \leq)$ and $(\mathbb{Q}, \leq)$ are NOT isomorphic. (where $\mathbb{N}$ is the set of natural numbers, $\mathbb{Q}$ is the set of rational numbers, $\leq$ is the usual 'less or equal to' between numbers).

Solution.

1. Omitted
2. Hint: Always map the minimal/least element in one structure to the other.
3. Suppose there is such an isomorphism function $f : \mathbb{N} \rightarrow \mathbb{Q}$. $f(0) = a$, $f(1) = b$. We have $a < b$ for $0 < 1$. Then consider $f^{-1}(\frac{a+b}{2})$. $\square$

Problem 2. Prove or disprove: If a partially ordered set $(X, \leq)$ has a single minimal element, then it is a smallest element as well.

Solution. Wrong. Consider $(\{a\}, \langle a, a \rangle) \cup (\mathbb{Z}, \leq)$. $\square$

Problem 3. Let $(X, \leq)$ and $(X', \leq')$ be partially ordered sets. A mapping $f : X \rightarrow X'$ is called an embedding of $(X, \leq)$ into $(X', \leq')$ if the following conditions hold:

- f is an injective mapping;
- $f(x) \leq' f(y)$ if and only if $x \leq y$.

Now consider the following problem

- 1) Describe an embedding of the set $\{1, 2\} \times \mathbb{N}$ with the lexicographic ordering into the ordered set $(\mathbb{Q}, \leq)$.
- 2) Solve the analog of a) with the set $\mathbb{N} \times \mathbb{N}$ (ordered lexicographically) instead of $\{1, 2\} \times \mathbb{N}$.

Solution.

- 1) $f(i, n) = i + y$ where $i \in \{1, 2\}$ and $y = \frac{n}{n+1}$.
- 2) Similarly.

□

Problem 4. Prove the following strengthening of the Erdős-Szekeres Lemma: Let κ, ℓ be natural numbers. Then every sequence of real numbers of length $\kappa\ell + 1$ contains an nondecreasing subsequence of length $\kappa + 1$ or a decreasing subsequence of length $\ell + 1$.

Solution. Hint: $\alpha(P) \cdot \omega(P) > \kappa\ell$. Then similar to the proof of Erdős-Szekeres Lemma: either $\omega(P) > \kappa$, which implies the existence of a nondecreasing subsequence of length $\kappa + 1$, or $\omega(P) \leq \kappa$, then $\alpha(P) > \ell$, which implies the other case. □

Problem 5. $I_1, I_2, \dots, I_{mn+1}$ are closed intervals on the real line. i.e., $I_j = [a_j, b_j]$ where $a_j \leq b_j$ for $1 \leq j \leq mn + 1$. Prove that:

- 1) either there are $m + 1$ intervals that are pair-wise disjoint
- 2) or, there are $n + 1$ intervals with a non-empty intersection.

[Hint: Dilworth's theorem]

Solution. Define a partial ordering $\sqsubseteq$ on the set of given intervals, where the strict relation is defined as:

$$I_r \sqsubset I_s \text{ iff } b_r < a_s.$$

Suppose that $I_{i_1}, I_{i_2}, \dots, I_{i_t}$ is a collection of pair-wise disjoint intervals. Assume that $a_{i_1} < a_{i_2} < \dots < a_{i_t}$. Then $I_{i_1} \sqsubset I_{i_2} \sqsubset \dots \sqsubset I_{i_t}$ form a chain. Conversely, a chain of length t comes from a set of t pair-wise disjoint intervals.

So, if 1) does not hold, then the maximum length of a chain is m . In this case the minimum number of chains needed to cover the poset is at least $\left\lceil \frac{mn+1}{m} \right\rceil = n + 1$. (i.e, lower bound of the number of chain partition is $n + 1$.)

Then by Dilworth's theorem, we can conclude that there must exist an anti-chain of length at least $n + 1$, denoted by $I_{j_1}, I_{j_2}, \dots, I_{j_{n+1}}$.

Let $a = \max \{a_{j_1}, a_{j_2}, \dots, a_{j_{n+1}}\}$ and $b = \min \{b_{j_1}, b_{j_2}, \dots, b_{j_{n+1}}\}$.

We must have $a \leq b$ for otherwise the two intervals giving a, b are disjoint. But then every interval of the anti-chain contains the non-empty fragment $[a, b]$, which gives the case 2). □